TMS / SYSTEM OVERVIEW

Power and protection

4S electronics power
5 V + analog branch
Independent RF 3.3 V

File: power.kicad_sch

12–16.8 V input –> 4.982 V nominal: 1 A plan.
Teensy +3V3: ADC logic. +3V3_RF: radio only.

Thrust acquisition

LC_MOSI
LC_MISO
LC_SCK
LC_CS_N
LC_DRDY_N

Bridge + buffers + ADS1256

LC_VREF

File: thrust.kicad_sch

Ratiometric reference

LC_VREF

VEXC/2 buffer

File: reference.kicad_sch

Reference ~2.49 V
Tracks +5VA excitation.

Teensy controller

Teensy 4.1
SD logging
Split SPI + UART

LC_MOSI
LC_MISO
LC_SCK
LC_CS_N
LC_DRDY_N
P_MOSI
P_MISO
P_SCK
P_CS_N
P_DRDY_N
XB_TX
XB_RX
XB_CTS_N
XB_RTS_N
FIRE_CMD

File: controller.kicad_sch

Pressure acquisition

P_MOSI
P_MISO
P_SCK
P_CS_N
P_DRDY_N

Pressure buffer + ADS1220

File: pressure.kicad_sch

XBee telemetry

XB_TX
XB_RX
XB_CTS_N
XB_RTS_N

Isolated UART / RP-SMA

File: radio.kicad_sch

Ignition interface

FIRE_CMD

2S + ARM + relay

File: ignition.kicad_sch

TARGETS: 1 kHz per channel, SD primary: telemetry 60 Hz / 32 bytes / 115200 baud.
A2 2026-10-14: datasheet corrections, exact part selection, buffered pressure, power-off UART isolation.
Read the verification guide for value calculations, design assumptions and remaining qualification checks.

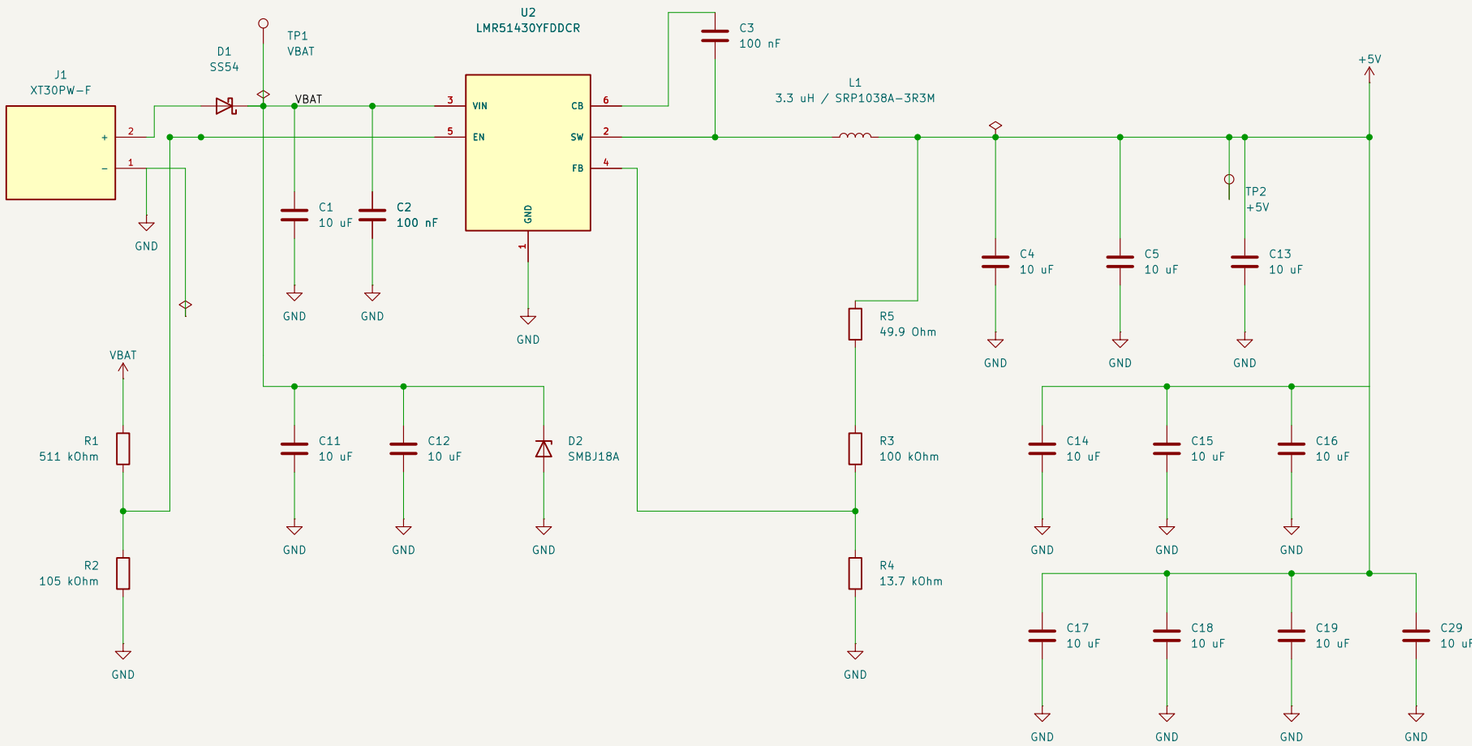
Engineering draft – see verification guide
Datasheet audit A2 – see verification guide
TMS – stand instrumentation

Sheet: /
File: tms.kicad_sch

Title: TMS / System overview

Size: A3	Date: 2026-10-14	**Rev: A2**
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POWER / 4S INPUT TO 5 V

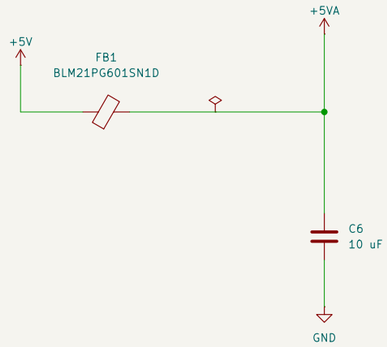


5 V SETPOINT AND LOAD

VOUT = 0.600 x (1 + 100.0499k / 13.7k) = 4.982 V.
Planning load: 1 A. IC limit is not a board current rating.
C3 is the floating bootstrap capacitor. R3/R4 set VOUT;
R5 permits loop-injection testing; C9 removed.
LMR51430YF: 1.1 MHz FPWM; L1 = 3.3 uH.
Input bank: C1/C11/C12 = 30 uF nominal, 50 V.
Output bank: C4/C5/C13-C19/C29 = 100 uF nominal, 50 V.
Use effective capacitance and the candidate audit bounds.

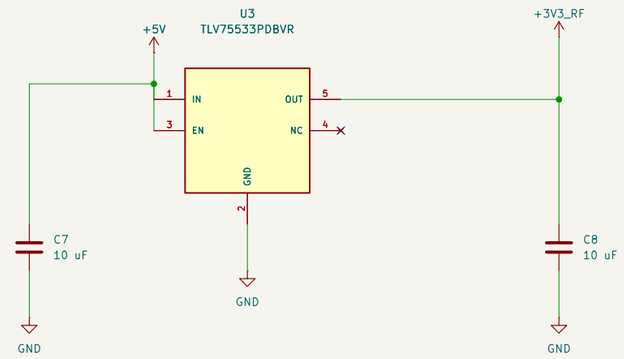
Input: 12-16.8 V. External fuse and 4S low-voltage
protection required. R1/R2 EN network is not a LiPo cutoff.
D2: 18 V stand-off; 29.2 V clamp at specified pulse.
See verification guide / TI LMR51430 Table 9-2.

ANALOG BRANCH



FB1 + C6 isolate the analog branch.
+5VA also excites the bridge (~13 mA).
FB1 DCR <=0.14 Ohm initial; verify PCB noise.

RADIO SUPPLY



3.3 V / 100 mA planning load.
LDO dissipation ~0.17 W.
Keep +3V3_RF separate from Teensy +3V3.

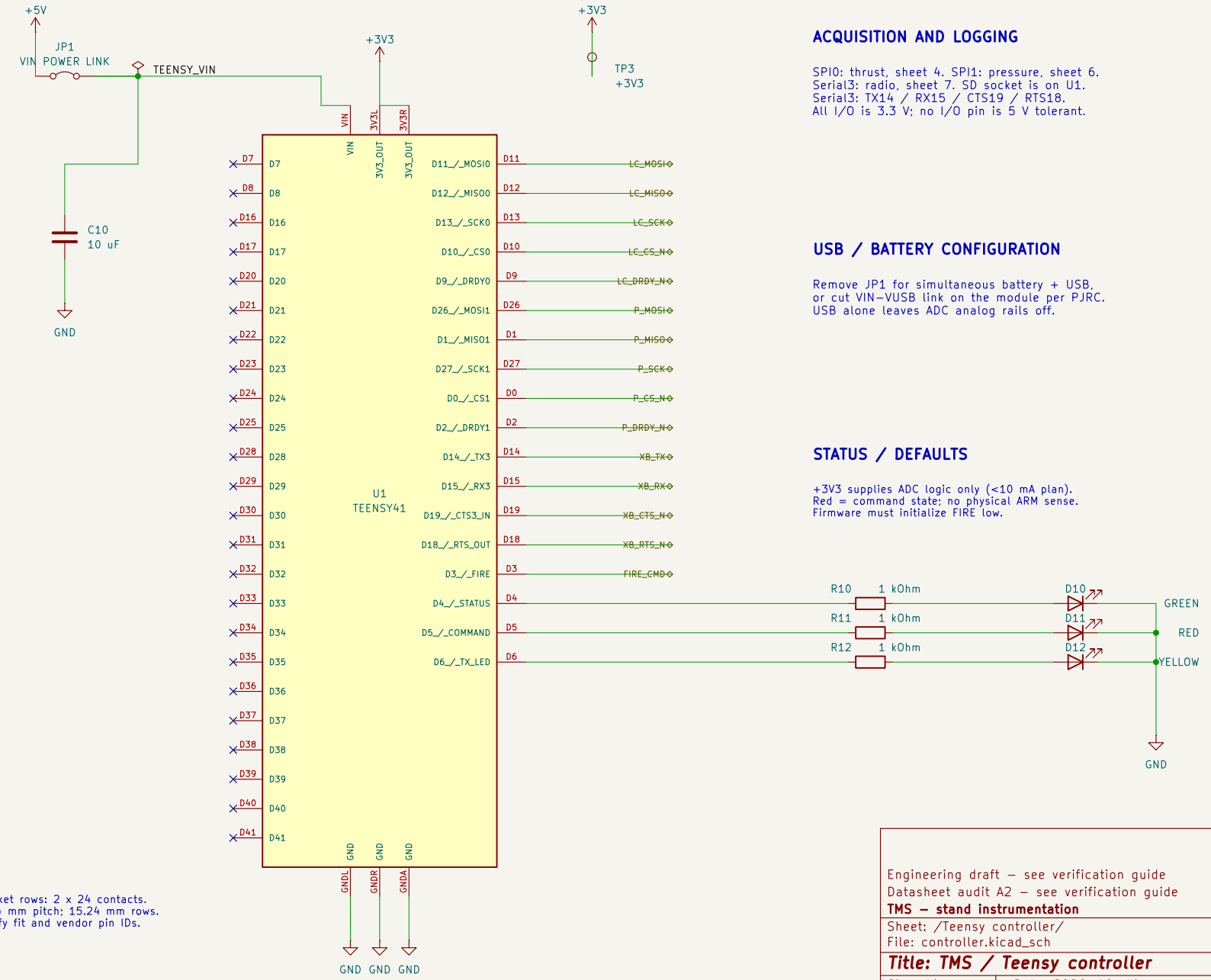
Engineering draft - see verification guide
Datasheet audit A2 - see verification guide
TMS - stand instrumentation

Sheet: /Power and protection/
File: power.kicad_sch

Title: TMS / Power and protection

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CONTROLLER / TEENSY 4.1



ACQUISITION AND LOGGING

SPI0: thrust, sheet 4. SPI1: pressure, sheet 6.
Serial3: radio, sheet 7. SD socket is on U1.
Serial3: TX14 / RX15 / CTS19 / RTS18.
All I/O is 3.3 V; no I/O pin is 5 V tolerant.

USB / BATTERY CONFIGURATION

Remove JP1 for simultaneous battery + USB,
or cut VIN–VUSB link on the module per PJRC.
USB alone leaves ADC analog rails off.

STATUS / DEFAULTS

+3V3 supplies ADC logic only (<10 mA plan).
Red = command state; no physical ARM sense.
Firmware must initialize FIRE low.

Socket rows: 2 x 24 contacts.
2.54 mm pitch; 15.24 mm rows.
Verify fit and vendor pin IDs.

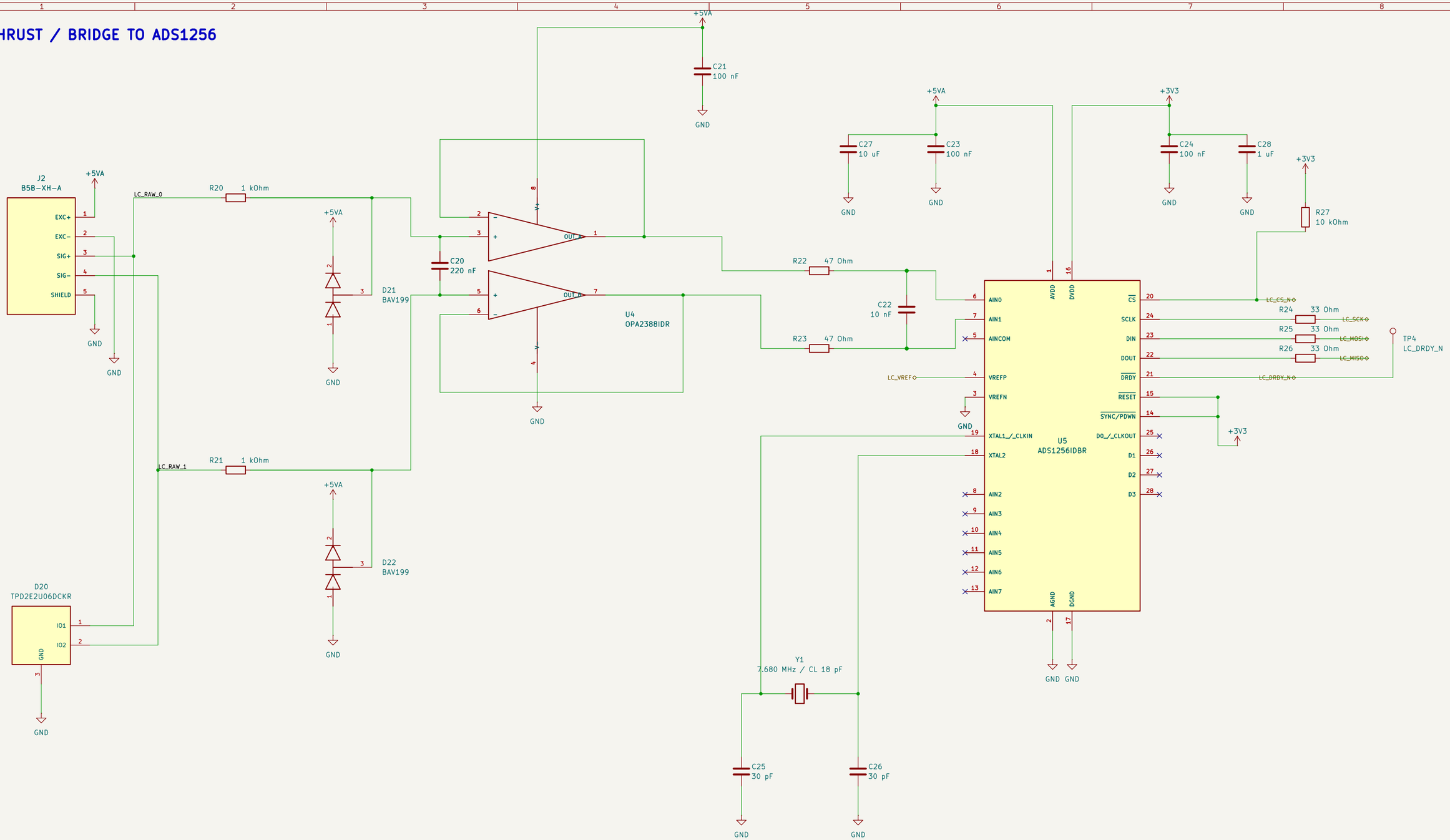
Engineering draft – see verification guide
Datasheet audit A2 – see verification guide
TMS – stand instrumentation

Sheet: /Teensy controller/
File: controller.kicad_sch

Title: TMS / Teensy controller

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THRUST / BRIDGE TO ADS1256



INPUT AND BUFFER

BCA-100L: nominal 10 mV at 100 kgf with 5 V excitation.
R20/R21 + C20: differential pole ~308 Hz including sensor.
U4 unity buffers avoid loading the bridge with the ADC.
Calibrate buffer offset, ADC loading and the complete chain.

ADC CONFIGURATION

AIN0-AIN1: PGA64; buffer ON.
1000 SPS; mode 1; SCLK <=1.92 MHz.
VREF ~2.49 V; differential FS ~+/-77.8 mV.
CL estimate: 30p/2 + 3p stray = 18 pF.

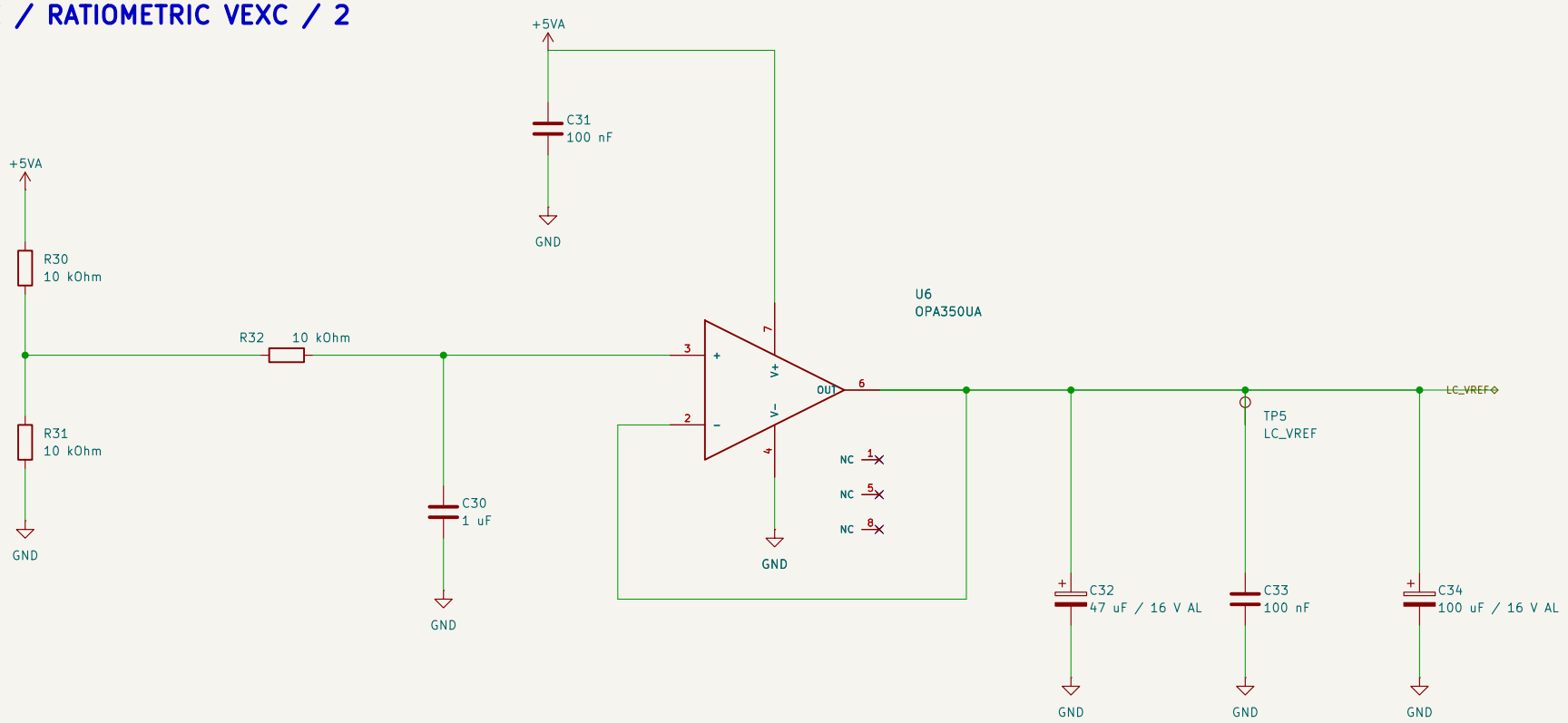
Engineering draft — see verification guide
Datasheet audit A2 — see verification guide
TMS — stand instrumentation

Sheet: /Thrust acquisition/
File: thrust.kicad_sch

Title: TMS / Thrust acquisition

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REFERENCE / RATIOMETRIC VEXC / 2



WHY THIS CIRCUIT

R30/R31 sense the actual bridge excitation. U6 drives ADC reference current.
VREF = VEXC x R31 / (R30 + R31) ~= VEXC/2, nominal 2.49 V.
R32/C30: (5k + 10k) x 1 uF = 15 ms; allow >=150 ms before calibration.
C32/C34: Panasonic FT; ESR max0.85/0.36 Ohm at100kHz. Validate loop stability.

Place C31 at U6 and the output bank at U5 VREFP/VREFN.
Unused package pads 1/5/8 are shown NC. See guide for ratio/offset and noise checks.

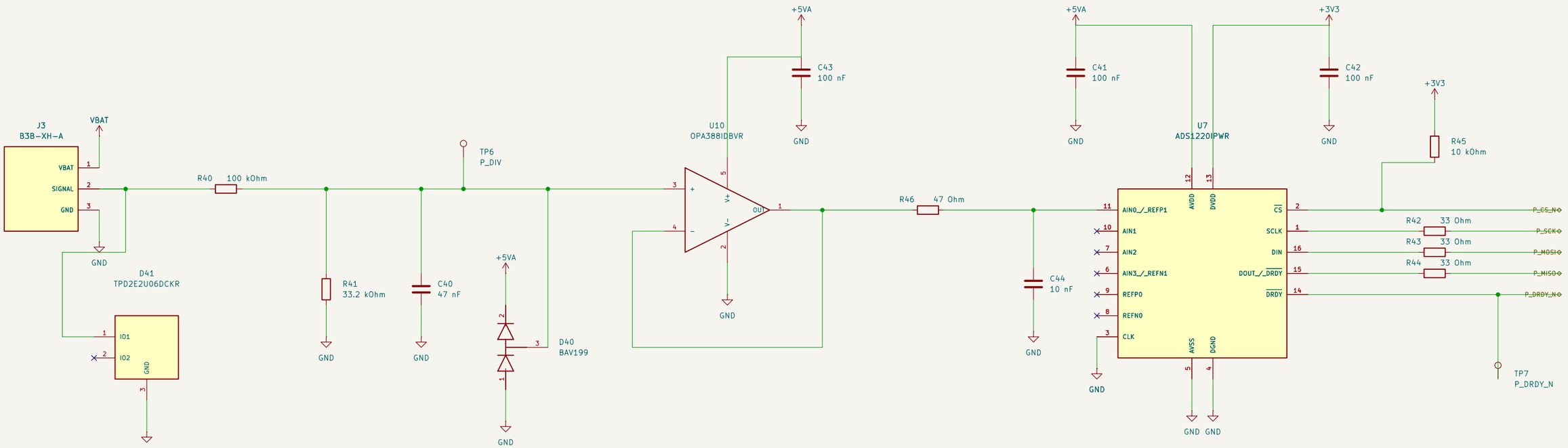
Engineering draft — see verification guide
Datasheet audit A2 — see verification guide
TMS — stand instrumentation

Sheet: /Ratiometric reference/
File: reference.kicad_sch

Title: TMS / Ratiometric reference

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PRESSURE / BUFFERED 0–5 V TO ADS1220



SCALING / 50 Hz DESIGN BAND

100k / 33.2k, 0.1%: k = 0.249249; 5 V -> 1.24625 V.
RTH = 24.925k; C40 = 47 nF C0G: fc ~=136 Hz.
U10 prevents the high-impedance divider from directly driving U7.
R46 / C44 isolate the op-amp from ADC charge pulses (47 Ohm / 10 nF).

ADC / FIRMWARE CONTRACT

AIN0-to-AVSS; gain1; PGA bypass; internal2.048 V.
Internal clock, normal1000 SPS; -61% positive range.
Unused analog/reference pins6/7/8/9/10: NC.
CLK3, DGND4, AVSS5: GND. IDAC / low-side switch OFF.

DESIGN ASSUMPTIONS / HARNESS

Sensor output resistance <=100 Ohm is assumed, not specified by Sensata. Calibrate the complete chain.
Board J3: supply / signal / return; sensor DIN: supply / return / signal (pins1/3/2).
Input is normally0-5 V. The sensor 32 V supply rating is not this board input rating.
Cable ESD, sustained miswiring and power-off injection need prototype qualification.

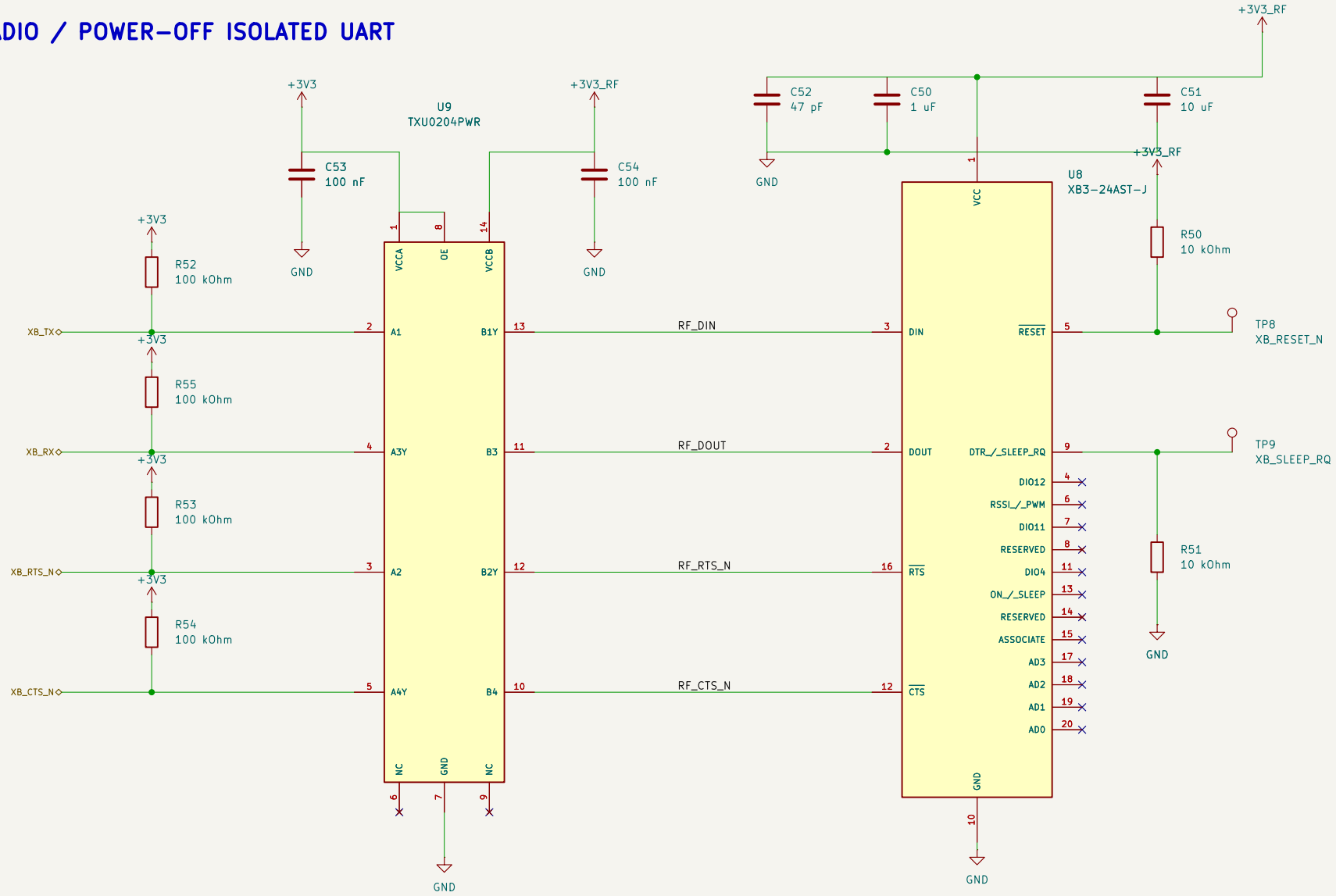
Engineering draft – see verification guide
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TMS – stand instrumentation

Sheet: /Pressure acquisition/
File: pressure.kicad_sch

Title: TMS / Pressure acquisition

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RADIO / POWER-OFF ISOLATED UART



U9 supplies: A = Teensy 3.3 V; B = RF 3.3 V. OE tied to VCCA.
Either rail off: U9 outputs Hi-Z. Common ground.
100k pulls define MCU idle and deassert RTS / CTS.

C52 / 47 pF closest to VCC. Ripple goal <=50 mV p-p. UART115200.
RP-SMA: XB3-24AST-J + A24-HASM-450.
USB adapter assumed at ground station.
Specify SoC Rev C+; inspect label per Digi p101.

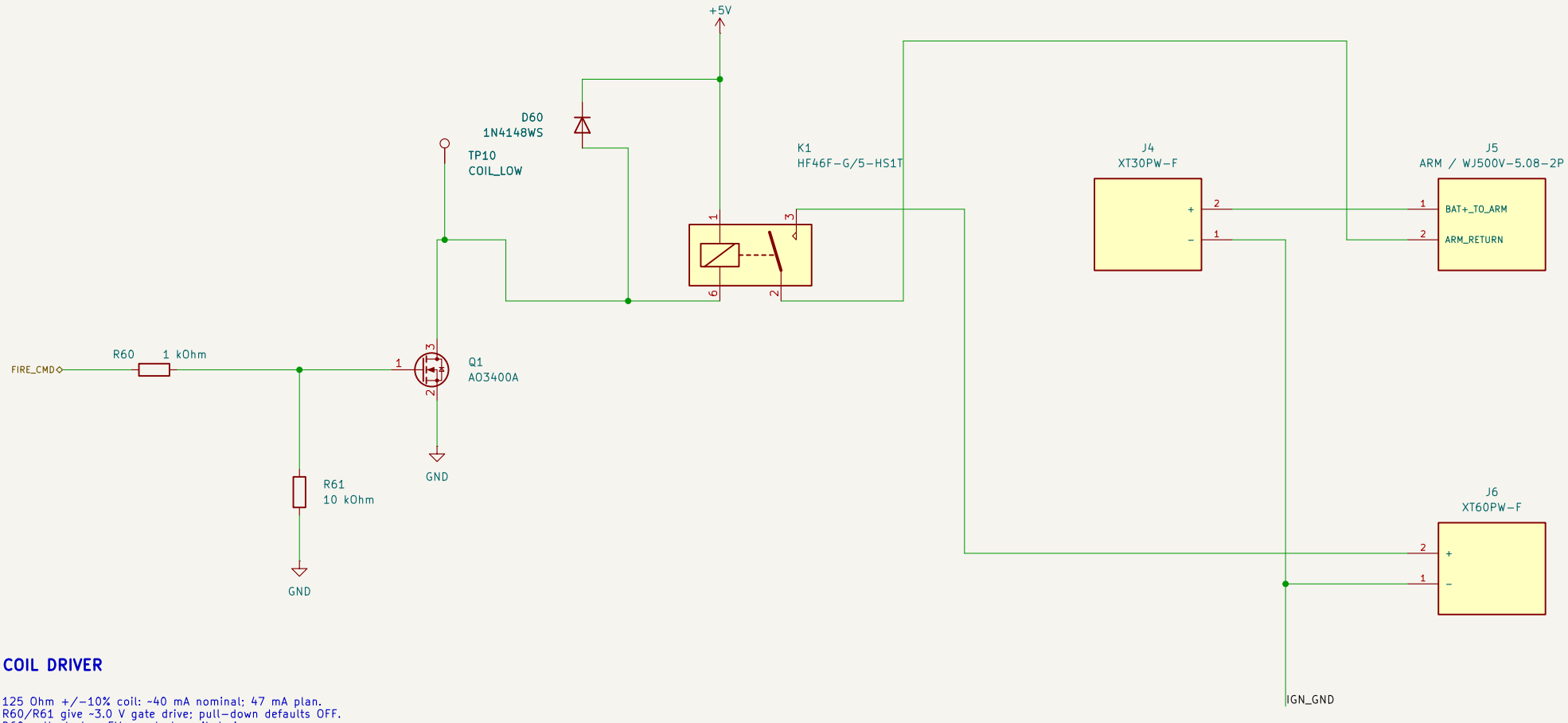
Engineering draft — see verification guide
Datasheet audit A2 — see verification guide
TMS — stand instrumentation

Sheet: /XBee telemetry/
File: radio.kicad_sch

Title: TMS / XBee telemetry

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IGNITION / COIL CONTROL + SERIES ARM



COIL DRIVER

125 Ohm +/-10% coil: ~40 mA nominal; 47 mA plan.
R60/R61 give ~3.0 V gate drive; pull-down defaults OFF.
D60 cathode to +5V; anode to coil drain.
Only the coil shares electronics GND.

SEPARATE CONTACT DOMAIN

2S -> physical ARM -> NO contact -> load.
IGN_GND has no electronics GND bond.
No continuity sense or physical ARM feedback.

Design assumption: load <=2 A at8.4 V; cold resistance >=4.2 Ohm.
Relay rated5 A resistive/30 VDC; assembled path still needs dummy-load qualification.
Use dummy-load reset/fault tests; no live ignition approval or two-fault inhibit claim.

Engineering draft - see verification guide
Datasheet audit A2 - see verification guide
TMS - stand instrumentation

Sheet: /Ignition interface/
File: ignition.kicad_sch

Title: TMS / Ignition interface

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